provided answers enables the system to learn from real interactions, improving its accuracy and user satisfaction over time.

5.5 Going deeper into LLM techniques

Here we summarize the key research that is revolutionizing how we interact with and evaluate LLMs. If you're focused on practical application rather than theoretical depth, you can skip this section.

5.5.1 Tree of thoughts

The *tree-of-thoughts* framework is an advanced prompting technique that improves an LLM's reasoning abilities by simulating human problem solving. Unlike the chain-of-thought method, which follows a single linear path, tree of thoughts enables the model to explore multiple reasoning paths in a tree-like structure. This approach is particularly effective for complex tasks that require strategic planning or multistep lookahead, such as solving mathematical problems or creative writing.

For more information, see the following:

- "Tree of Thoughts: Deliberate Problem Solving with Large Language Models" by Yao et al., https://arxiv.org/abs/2305.10601
- Tree of Thoughts (ToT), https://www.promptingguide.ai/techniques/tot

5.5.2 Self-refine

Self-refine is an innovative approach that allows an LLM to iteratively improve its own outputs without the need for additional training data or reinforcement learning. The process works in a feedback loop: an LLM generates an initial output and then provides itself with feedback on how to improve it. Based on this feedback, the model refines its original output. This cycle can be repeated until the output meets a desired quality standard. This technique is highly effective for tasks with complex or hard-to-define goals, such as review rewriting or code optimization.

For more information, see the following:

- "Self-Refine: Iterative Refinement with Self-Feedback" by Madaan et al., https://arxiv.org/abs/2303.17651
- Project page, https://github.com/madaan/self-refine

5.5.3 LLM-as-a-judge

The *LLM-as-a-judge* technique uses the power of LLMs to evaluate the quality of text generated by other LLMs. This is an alternative to traditional human evaluation, which can be costly and time consuming. By providing a "judge" LLM with a set of guidelines, it can assess various aspects of a generated text, such as correctness, tone, or conciseness. While this method offers advantages in efficiency and consistency, potential biases in the judge LLM might occur; careful prompt design is essential to ensure reliability.